Mathematics Class 11th

Solved Model Paper 2026

Smart Syllabus Punjab Boards

Q.1 MCQS

i)

$(1,0)$ Mult.

$x+yi$

$1+0i = 1$

Real

$1 \times 2 = 2$

$1 \times 3 = 3$

$1 \times 0 = 0$

$5 \times \frac{1}{5} = 1$

$z \times z^{-1} = 1+0i = \boxed{(1,0)}$

$\sqrt{4-5} = \sqrt{-1}$ $\sqrt{4-\square}$

$\sqrt{0} = 0$ $\sqrt{4-9}$

iii. The domain of the function $f(x)=\sqrt{4-x^2}$ is:
(A) $(-2, 2)$ (B) $(-2, \infty)$ (C) $(-\infty, 2)$ (D) $[-2, 2]$

$4-x^2=0 \Rightarrow x^2 = 4 = (2)^2$

$(2-x)(2+x)=0$

$x=2, x=-2$

$x = \pm 2$

$-\infty$ -2 $+2$ $+\infty$

-3

$\sqrt{-ve}$

(-1) $(-2, 2)$

$2 < 5$

$-2 > -5$

$\sqrt{x^2} = |x|$

$\begin{cases} +x, & x \geq 0 \\ -x, & x < 0 \end{cases}$

$4 - x^2 \geq 0$

$-x^2 \geq -4$

$\sqrt{x^2} \leq \sqrt{4}$

$\pm x \leq 2$

$\boxed{x \leq 2}$

$-x \leq 2$

$\boxed{x \geq -2}$

$\boxed{x \leq \cancel{\pm 2}}$

-5 -2

$-2 \leq x \leq 2$

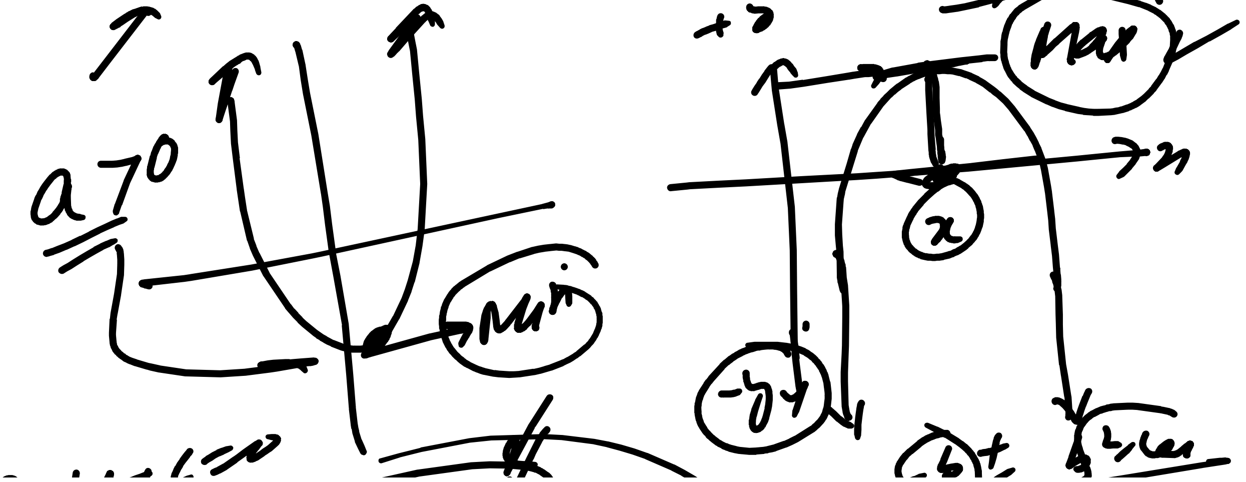
$[-2, +2]$

$x = -\frac{b}{2a} =$

iv.	What is the maximum value of $f(x) = -2x^2 + 4x + 3$?
	(A) 1 (B) 3 (C) 5 (D) 7

$ax^2 + bx + c$ -ve

Parabola



$ax^2+bx+c=0$

$\frac{-b \pm \sqrt{b^2-4ac}}{2a}$

$f(x) = -2x^2+4x+3$

$\boxed{x = -\frac{b}{2a}} = -\frac{4}{2(-2)} = \frac{-4}{-4} = +1$

MAX

$f(1) = -2+4+3 = 2+3 = 5$

$f'(x) = -2(2x+4) = 0$

$\Rightarrow -4x+4=0$

$4x=4$

$x=1$

x^2

x^1 $1-1$

$1x = 1x^0$

xx.	The magnitude of a vector perpendicular to both the vectors $u = 2i - 3j + k$ and $v = i + 4j - 2k$ is:							
	(A)	$\sqrt{50}$	(B)	$\sqrt{150}$	(C)	$\sqrt{500}$	(D)	$\sqrt{1550}$

$|\vec{U} \times \vec{V}| = \begin{vmatrix} i & j & k \\ 2 & -3 & 1 \\ 1 & 4 & -2 \end{vmatrix} = i(6-4) - j(-4-1) + k(8+3)$

$= 2i + 5j + 11k$

$\sqrt{2^2+5^2+11^2} = \sqrt{4+25+121}$

Example

$\Rightarrow$ EX 1.1

Subjective Type (Part I)

EX 1.1

Subjective Type (Part-I)

Time allowed: 2:30 hours **Max. Marks:80**

2. Write short answers to any EIGHT (8) questions: **$2 \times 8 = 16$**

(i) If $z_1 = (4, 2)$ and $z_2 = (3, -1)$, then find $\frac{z_1}{z_2}$.

$z_1 = 4+2i$, $z_2 = 3-i$

$$\frac{z_1}{z_2} = \frac{4+2i}{3-i} = \frac{4+2i}{3-i} \times \frac{3+i}{3+i}$$

$$= \frac{(4+2i)(3+i)}{(3-i)(3+i)}$$

$$= \frac{4(3+i)+2i(3+i)}{3^2-i^2}$$

$2(-1)$

-2

$$= \frac{12+4i+6i+2i^2}{9+1}$$

$$= \frac{12-2+10i}{10} = \frac{10+10i}{10}$$

$$= \frac{\cancel{10}(1+i)}{\cancel{10}} = 1+i$$

$$\boxed{\frac{z_1}{z_2} = 1+2i}$$

EX # 1.1 Q 7

Set of Nat

(ii) Show that: $i^{n+1} + i^{n+2} + i^{n+3} + i^{n+4} = 0$, for all $n \in N$.

$n = \{1, 2, 3, \ldots\}$

$L.H.S = i^{n+1} + i^{n+2} + i^{n+3} + i^{n+4}$

$(n+1)+1$

$(n+1)+2$

$(n+1)+3$

$= i^{n+1}(1 + i + i^2 + i^3)$

$= i^{n+1}(\cancel{1} + \cancel{i} - \cancel{1} - \cancel{i})$

$i^3 = i \cdot i^2$

$= -i$

$= i^{n+1} \times 0 = 0$ As

Example # 5 → EX # 1.2

(iii) Find the square root of complex number $5 + 12i$.

Let $Z = 5 + 12i = x + iy$

$x = 5, \quad y = 12, \quad y > 0$ (+ve)

$$\sqrt{x+yi} = \pm\left(\sqrt{\frac{|z|+x}{2}} + i\frac{y}{|y|}\sqrt{\frac{|z|-x}{2}}\right)$$

$z = 5 + 12i$, $|z| = \sqrt{5^2 + (12)^2} = \sqrt{25 + 144} = \sqrt{169}$
$= 13$

$\boxed{|z| = 13}$

$$\sqrt{5+12i} = \pm\left(\sqrt{\frac{13+5}{2}} + i\,\frac{12}{12}\sqrt{\frac{13-5}{2}}\right)$$

$$= \pm\left(\sqrt{\frac{18}{2}} + i\sqrt{\frac{8}{2}}\right)$$

$$\sqrt{5+12i} = \pm(3 + 2i)$$

The sq. Rot of $5+12i$ is $\boxed{\begin{matrix} 3+2i \\ -3-2i \end{matrix}}$ Ans

$$\sqrt{5+12i} = \pm(x+yi) \rightarrow (A)$$

$$5+12i = (x+yi)^2$$

$$5+12i = x^2 - y^2 + 2xyi$$

x y
6x1
3x2

$\boxed{x^2 - y^2 = 5}$

$2xy = 12 \Rightarrow \boxed{xy = 6}$

9-4

$x = 3$

9-4

$x = 3$

$y = 2$

$$\sqrt{5+12i} = \boxed{\pm(3+2i)}$$

Ex#1.4 → Q.8

(iv) If ω is an imaginary cube roots of unity, prove that $\dfrac{a+b\omega^2+c\omega}{a\omega^2+b\omega+c}=\omega$

$1+\omega+\omega^2=0$

$\omega^3 = 1$

$$\text{L.H.S} = \frac{a+b\omega^2+c\omega}{a\omega^2+b\omega+c}$$

$$= \frac{\omega^2}{\omega^2}\left(\frac{a+b\omega^2+c\omega}{a\omega^2+b\omega+c}\right)$$

$\omega^4 = \omega^3\cdot\omega$ ①

$= \omega$

$$= \frac{1}{\omega^2}\left(\frac{a\omega^2+b\omega^4+c\omega^3}{a\omega^2+b\omega+c}\right)$$

$$= \frac{1}{\omega^2}\left(\frac{\cancel{a\omega^2+b\omega+c}}{\cancel{a\omega^2+b\omega+c}}\right)$$

$$= \frac{1}{\omega^2} = \frac{\omega}{\omega^3} = \frac{\omega}{1} = \omega$$

$= \text{RHS}$

$\Rightarrow$ LHS = RHS Hence proved

Ex # 2.1 Q5

(v) Given $f(x) = x^3 - ax^2 + bx + 1$. If $f(2) = -3$ and $f(-1) = 0$. Find the values of a and b.

$$f(2) = -3$$

$$8 - 4a + 2b + 1 = -3$$

$$9 - 4a + 2b = -3$$

$$-4a + 2b = -12$$

$$-2a + b = -6$$

$$2a - b = 6 \rightarrow (1)$$

Also, $f(-1) = 0$

$$\cancel{-1} - a - b + \cancel{1} = 0$$

$$-a - b = 0$$

$$\Rightarrow \boxed{a + b = 0} \rightarrow (2)$$

put $\boxed{a = -b}$ in (1)

$$2(-b) - b = 6$$

$$2(-b) - b = 6$$
$$-2b - b = 6$$
$$-3b = 6$$
$$\boxed{b = -2}$$
$$a = -(-2) \Rightarrow \boxed{a = 2}$$
Thus $\boxed{a = 2}$ & $\boxed{b = -2}$

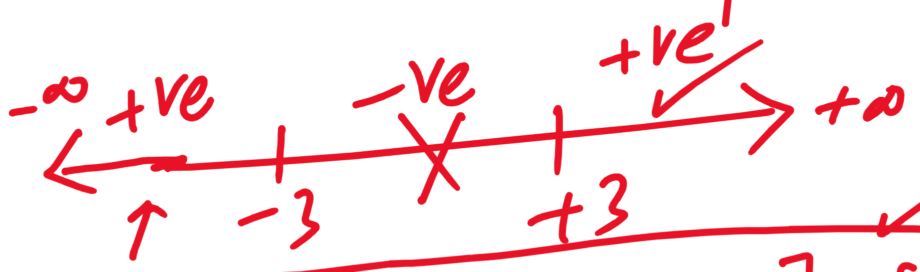
Ex # 2.1 Example # 4

$x^2 - 9 \geq 0$

(vi) Find the domain and range of $f(x) = \sqrt{x^2 - 9}$

$\sqrt{0} = 0$ ✓

$$x^2 - 9 \geq 0 \qquad (x)^2 - (3)^3 = 0$$
$$(x-3)(x+3) = 0$$
$$x - 3 = 0 \qquad x + 3 = 0$$
$$x = 3 \qquad x = -3$$



$$\boxed{\text{Dom } f(x) = (-\infty, -3] \cup [3, +\infty)}$$
$$\boxed{\text{Rang } f(x) = [0, +\infty)}$$

2nd method

$\sqrt{x^2} = |x|$

$x^2 - 9 \geq 0$

$\sqrt{x^2} \geq \sqrt{9}$

$\pm x \geq 3$

Case-1

$x \geq 3$

$[3, +\infty)$

Cas

$-x \geq 3$

$\boxed{x \leq -3}$

$(-\infty, -3]$

Ex#3.1 Q.3(iv) $y = f(x)$

(vii) Find the inverse of $f(x) = 3x^2 - 2x + 6,\ x \geq 5.$

Sol

$$y = 3x^2 - 2x + 6$$

Replace x by y & y by x

$$x = 3y^2 - 2y + 6$$

$$3y^2 - 2y + (6 - x) = 0$$

$$y = \frac{2 \pm \sqrt{4 - 4(3)(6-x)}}{6}$$

$$y = 2 \pm 2\sqrt{1 - 18 + 3x}$$

$$y = \frac{2 \pm 2\sqrt{1-18+3x}}{6}$$

$$y = \frac{2\left(1 \pm \sqrt{3x-17}\right)}{6}$$

$$\boxed{y = \frac{1 \pm \sqrt{3x-17}}{3}}$$

$x \geq 5$

$\text{Dom} f = [5, +\infty)$

$y \geq 5$

$$y = \frac{1+\sqrt{3x-17}}{3} \qquad y = \frac{1-\sqrt{3x-17}}{3}$$

X

it does satisfy the given (?)

$\boxed{\text{Rang } f = [5, +\infty)}$

$f^{-1}(x)$ $\qquad f^{-1}(x) =$

$y = f(x)$

$x = f^{-1}(y)$

$\frac{1-0}{3}$

$$\frac{1+14}{3} = 5$$

$$\frac{1+\sqrt{3x-17}}{3} \geq 5$$

$$\sqrt{3x-17} = 15-1 \geq 14$$

$$3x-17 \geq 196$$

$$3x \geq 213 \Rightarrow \boxed{x \geq 71}$$

$$\boxed{\text{Dom} f^{-1} = [71, +\infty)}$$

$\text{Dom } f^{-1} = [1, +\infty)$

Ans $\boxed{f^{-1}(x) = 1 + \frac{\sqrt{3x - 17}}{3}}$

$\text{Range } f^{-1} = [3, +\infty)$

Ex # 3.1 Q.4 (i)

$\leq, \geq$

(viii) Solve $|x^2 + 1| = 5$

$\pm (x^2+1) = \pm 5$

$x^2 + 1 = 5$

$x^2 - (2)^2 = 0$

$x^2 = 4$

$x = \pm 2$

$x^2 + 1 = -5$

$x^2 = -6$

Not possible

$\boxed{S.S = \{-2, +2\}}$

Ex # 4.1 Q4 (i)

(ix) If A and B are square matrices of the same order, then explain why in general $(A+B)^2 \neq A^2 + 2AB + B^2$.

$LHS = (A+B)^2 =$

$= (A+B)(A+B)$

$$= (A+B)(A+B)$$

$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $$= A(A+B) + B(A+B)$$

$B = \begin{bmatrix} -9 & 2 \\ 1 & -5 \end{bmatrix}$ $$= A^2 + \underline{\underline{AB + BA}} + B^2$$

In matrices in general

AB $\begin{bmatrix} -9+2 & 2-10 \\ -27+4 & 6-20 \end{bmatrix}$ $$AB \neq BA$$ BA $\begin{bmatrix} -9+6 \\ \end{bmatrix}$

$\begin{bmatrix} -7 & -8 \\ -23 & -14 \end{bmatrix}$ $$AB + \underline{\underline{BA}} \neq 2AB$$

$$\neq \text{RHS.}$$

Thus $(A+B)^2 \neq A^2 + 2AB + B^2$.

Ex#4.2 Q(5) $\underline{\underline{i}}$

(x) Find the values of x if $\begin{vmatrix} 2 & 1 & x \\ -1 & -4 & -3 \\ x & 1 & 0 \end{vmatrix} = 5$

$$2(0-(-3)) - 1(0-(-3x)) + x(-1-(-4x)) = 5$$

$$6 - 3x + x(-1+4x) = 5$$

$$6 - 3x - x + 4x^2 = 5$$

$6-3x + x \quad 4x \quad 3$

$$\boxed{4x^2-4x+1=0}$$

$$x = \frac{4 \pm \sqrt{16-4(4)(1)}}{8}$$

$$x = \frac{4\pm 0}{8} \Rightarrow \boxed{x=\frac{1}{2}}$$

Thus $\boxed{x=\frac{1}{2}}$

k^2

Ex 4.2, Q7

(xi) If A is a square matrix of order 3, then show that $|kA| = k^3|A|$.

Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ ✓

$KA = \begin{bmatrix} Ka_{11} & Ka_{12} & Ka_{13} \\ Ka_{21} & Ka_{22} & Ka_{23} \\ Ka_{31} & Ka_{32} & Ka_{33} \end{bmatrix}$

$$|KA| = \begin{vmatrix} Ka_{11} & Ka_{12} & Ka_{13} \\ Ka_{21} & Ka_{22} & Ka_{23} \\ Ka_{31} & Ka_{32} & Ka_{33} \end{vmatrix}$$

Taking "K" common from 1^{st} Row

$$|KA| = K \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ Ka_{21} & Ka_{22} & Ka_{23} \\ Ka_{31} & Ka_{32} & Ka_{33} \end{vmatrix}$$

Taking "K" - —— R_2

$$= (K)(K) \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ Ka_{31} & Ka_{32} & Ka_{33} \end{vmatrix}$$

Taking "K" - —— R_3

$$|KA| = (K)(K)(K) \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$|KA| = K^3|A| \rightarrow \text{Hence proved}$$

Ex 5.1 Q (i)

(xii) Resolve $\frac{2}{x^2-1}$ into partial fractions:

$\frac{A}{\;} + \frac{B}{\;} \rightarrow (1)$

$$\frac{2}{x^2+1} = \frac{A}{x-1} + \frac{B}{x+1} \rightarrow (1)$$

$\times$ by $(x-1)(x+1)$ on both sides, we get

$$2 = A(x+1) + B(x-1) \rightarrow (2)$$

put $x-1=0$ in Eqn. (2)

$\boxed{x=1}$

$$2 = A(1+1) \Rightarrow 2 = 2A$$

$\boxed{A=1}$

put $x+1=0$ in Eqn (2), we

$\boxed{x=-1}$

$$2 = B(-1-1)$$

$$-2B = 2$$

$\boxed{B=-1}$

$$\boxed{\frac{2}{x^2-1} = \frac{1}{x-1} - \frac{1}{x+1}}$$

$$= \frac{x+1-(x-1)}{(x+1)(x-1)} = \frac{\cancel{x}+1\cancel{-x}+1}{x^2-1}$$

Ex # 6.2 Q(8)

3. Write short answers to any EGHT (08) questions: $2 \times 8 = 16$

(i) Which term of the A.P. 2, 8, 13, ... is 123?

3. **Write short answers to any EGHT (08) questions:** **2 × 8 = 16**

(i) Which term of the A.P., 3, 8, 13, … is 123?

(ii) Find the sum of the first 100 positive integers.

6.9 (iii) Find the 12[th] term of $1 + i, 2i, -2 + 2i, \cdots$.

Q4 (iv) If 5 is the harmonic mean between 2 and b, find b.

(v) Evaluate $\frac{9!}{6!\,3!}$

6 Ex. 6.5 Q5

Example 7 Ex # 6.4

i) $3, 8, 13, \ldots, 123$

$a_1 \quad a_2 \quad a_3, \quad a_?$

Let $\boxed{a_n = 123}$ ✓ $n = ??$

✓ $a_1 = 3$, ✓ $a_2 = 8$

$d = a_2 - a_1 = 8 - 3 = 5$

General Term

$a_n = a_1 + (n-1)d$

$123 = 3 + (n-1)5$

$(n-1)5 = 120$

$n - 1 = 24$

$\boxed{n = 25}$ Ans

Thus $\boxed{a_{25} = 123}$

Verification $a_{25} = a + 24d$

$= 3 + 24(5) = 3 + 120$ 2

Verify: $a_{25} = \ldots$
$= 3 + 24(5) = 3 + 120$

ii) $1 + 2 + 3 + \cdots + 100 \rightarrow a_n$

$S_n = \frac{n}{2}(a_1 + a_n)$, $\boxed{S_n = \frac{n}{2}[2a_1 + (n-1)d]}$

$a_1 = 1$, $a_2 = 2$, $n = 100$

$S_{100} = \frac{100}{2}(1 + 100) = \frac{100}{2} \times 101$

5050

$S_{100} = \frac{100}{2}[2 + (100-1)1]$

$= 50(2 + 99) = 50 \times 101$

$= 5050$

(iii) Find the 12th term of $1 + i, 2i, -2 + 2i, \cdots$.

$n = 12$

$\boxed{a_n = a_1 r^{n-1}}$

General term of G.P.

$a_1 = 1 + i$
$a_2 = 2i$

$r = \frac{2i}{1+i} \rightarrow \frac{a_2}{a_1} = 1 + i$

$\frac{-2+2i}{\ldots} = 1 + i$

$a_2 = 2i$

$a_3 = -2+2i$, $r = \frac{a_3}{a_2} = \frac{-2+2i}{2i} = 1+i$ ✓

Given $n = 12$, $a_1 = 1+i$
$a_2 = 2i$

$a_{12} = ?$

Sol

$r = \frac{a_2}{a_1} = \frac{2i}{1+i} = 1+i$

$a_n = a_1 r^{n-1}$

$a_{12} = (1+i)(1+i)^{11}$

$\boxed{a_{12} = (1+i)^{12}}$

$r = \sqrt{2}$

$\theta = 45^\circ = \frac{\pi}{4}$

$\boxed{a_{12} = -64}$

$(x+iy)^n \Rightarrow (r\cos\theta + i\sin\theta)^{12}$

$(\sqrt{2}(\cos 45^\circ + i\sin 45^\circ))^{12}$

$= (\sqrt{2})^{12}(\cos 540 + i\sin 540)$

$= 64(-1 + \ldots)$

$(1+i)^{12} = -64$

6.9 → Q(4)

$2, H, b$

(iv) If 5 is the harmonic mean between 2 and b, find b.

H

(v) Evaluate $\dfrac{9!}{6!\,3!}$ → Example (3) 7.1

Given
$H = 5$
$a = 2$
$b = b?$

$$H = \frac{2ab}{a+b}$$

$$5 = \frac{2(2)(b)}{(2+b)}$$

$$10 + 5b = 4b$$

$$5b - 4b = -10$$

$$\boxed{b = -10}$$ Ans

$\boxed{2, 5, b}$

$\boxed{2, 5, -10}$

$$\frac{2(2)(-10)}{-8} = \frac{40}{8} = 5$$

v) $\frac{9!}{6!\cdot 3!}$ ≠

$= \frac{\cancel{9}^{3}\cdot \cancel{8}^{4}\cdot 7\cdot \cancel{6!}}{\cancel{6!}\cdot \cancel{3}\cdot \cancel{2}\cdot 1}$

$= 84$ Ans

$\boxed{84}$

$\boxed{n! = n(n-1)!}$ 3·2·1

$\boxed{6\cdot 5\cdot 4\cdot 3\cdot 2\cdot 1}$

6!

$\frac{12}{7}$

$\frac{9\cdot 8\cdot 7\cdot \cancel{6\cdot 5\cdot 4\cdot 3\cdot 2\cdot 1}}{\cancel{6\cdot 5\cdot 4\cdot 3\cdot 2\cdot 1}\cdot 3\cdot 2\cdot 1}$

7.2 Q6 12, 21 ab, ba order 7.4 Q11

ABC CAB BCA

(vi) How many signals can be given by 6 flags of different colours, using 2 flags at a time?

(vii) How any diagonals and triangles can be formed by joining the vertices of a the polygon having 15 sides?

1 2
R B
B R

R B O W Y P

2 1
2

n = 6
r = 2

Number

# of signal = $^6P_2 = \frac{6!}{(6-2)!}$

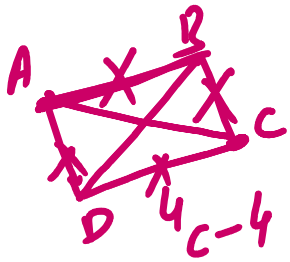
6! 6·5·4! = 30

$$= \frac{6!}{4!} = \frac{6\cdot5\cdot4!}{4!} = 30 \text{ sides}$$

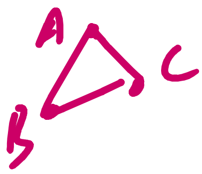
$n = 15 \rightarrow$ No. of sides

(vii) No of diagonals

$$= {}^{n}C_{2} - n$$



$${}^{4}C_{2} - 4 = \cancel{6-4}$$



$$= {}^{15}C_{2} - 15 = 90$$

No of Triangles $= {}^{15}C_{3} = \frac{15!}{3!\,12!}$

$$= \frac{15\cdot14\cdot13\cdot\cancel{12!}}{3\cdot2\cdot1\cdot\cancel{12!}}$$

AB, BA ✓

ABC, ACB ✓

$= 455$ triangles

$9.1 \rightarrow Q7$

Example 9 $\rightarrow 9.2$

(viii) When the polynomial $4x^4 + 2x^3 + kx^2 + 13$ is divided by $x + 1$, the remainder is 16. Find the value of k.

(ix) Suppose a polynomial regression model $P(x) = 3x^3 - 4x^2 + 2x - 5$. If a data point at $x = -1$ is missing. What should be its predicted value?

viii) Given $P(-1) = 16 \quad K = ?$

viii) Given $P(-1) = 16$ $K = ?$

where $P(x) = 4x^4 + 2x^3 + Kx^2 + 13$

$$P(-1) = 4(1) + 2(-1) + K + 13 = 16$$

$$\Rightarrow \quad 4 - 2 + K + 13 = 16$$

$$K + 15 = 16$$

$$\boxed{K = 1.}$$

Ans

ix) given $P(x) = 3x^3 - 4x^2 + 2x - 5$

To find $P(-1) = ?$

$$P(-1) = 3(-1) - 4(1) + 2(-1) - 5$$

$$P(-1) = -3 - 4 - 2 - 5$$

$$\boxed{P(-1) = -14}$$

Ans

$\boxed{P(-1) = -14}$ 5-1

Thus at $x = -1$, -14 was missing

10.1 Q3 (iv) 90°, 180, 270, 360

(x) Prove that : $\sin 210° + \cos 240° + \tan 225° + \cot 225° = 1$.

LHS $= \sin 210° + \cos 240° + \tan 225° + \cot 225°$

$= \sin(180° + 30°) + \cos(270° - 30°) + \tan(270° - 45°) + \cot(270° - 45°)$

$= -\sin 30° - \sin 30° + \cot 45° + \tan 45°$

$= -\frac{1}{2} - \frac{1}{2} + 1 + 1$

$= \cancel{-1} + \cancel{1} + 1 = 1 = \text{RHS}$

Hence proved

10.2 Q2(ii)

10.4 Example 16

(xi) Prove that: $\cos(\alpha + 45°) = \frac{1}{\sqrt{2}}(\cos \alpha - \sin \alpha)$.

(xii) Express $\sin 5x + \sin 7x$ as a product.

xi) $\boxed{\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta}$

$$LHS = \cos(\alpha+45^\circ) = \cos\alpha\sin45^\circ - \sin\alpha\cos45^\circ$$

$$= \cos\alpha\frac{1}{\sqrt{2}} - \sin\alpha\frac{1}{\sqrt{2}}$$

$$= \frac{1}{\sqrt{2}}(\cos\alpha - \sin\alpha)$$

Hence proved $= RHS$.

$\Rightarrow LHS = RHS$

xii) $\overset{\alpha}{\sin 5x} + \overset{\beta}{\sin 7x}$

$$\sin\alpha + \sin\beta = 2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$

$$= 2\sin\left(\frac{5x+7x}{2}\right)\cos\left(\frac{5x-7x}{2}\right)$$

$$= 2\sin 6x\cos(-x)$$

$\boxed{\cos(-x) = \cos x}$

$\sin 7x + \sin 5x$

$$\boxed{\sin 5x + \sin 7x = 2\sin 6x\cos x}$$

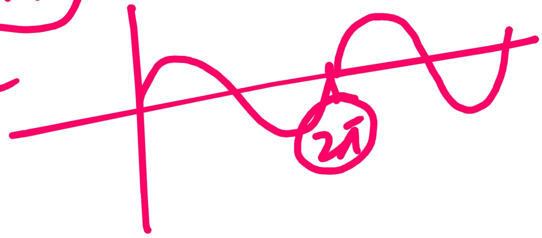
Qy $\frac{3x}{2} = 2\pi$ $3x = 4\pi$ $x = 4\pi/3$ 11.1 Q2 (vii)

(i) Find the periods of $\frac{1}{2}\sin\left(\frac{3x}{2}-\frac{\pi}{2}\right)$.

sinx, cosx, secx, cosec x $\rightarrow$ 2π

tan, cot $\rightarrow \pi$

$\sin(x+2\pi) = \sin x$



Sol For period, we consider $\sin\left(\frac{3x}{2}\right)$

As we know that period of sinx is 2π

$\therefore \sin(x + 2\pi) = \sin x$

So, $\sin\left(\frac{3x}{2}\right) = \sin\left(\frac{3x}{2} + 2\pi\right)$

$= \sin\left(\frac{3x}{2} + \frac{2\cdot 2\pi}{2}\right)$

$\sin\left(\frac{3}{2}\left(x + \frac{2\pi \cdot 2}{3}\right.\right.$

$= \sin\left(\frac{3x}{2} + \frac{3\cdot 4\pi}{3\cdot 2}\right)$

$= \sin\left(\frac{3}{2}\left(x + \frac{4\pi}{3}\right)\right)$

As sin 3x repeats its if x is

As $\sin\frac{3x}{2}$ repeats its if x is increased by $4\pi/3$ so,

$\boxed{\text{Period of } \sin\frac{3x}{2} \text{ is } \frac{4\pi}{3}}$

11.3 Q1(iv)

(ii) Find the maximum and minimum values of $\frac{3}{2}+\cos\left(x-\frac{\pi}{4}\right)$.

Max value $= \frac{3}{2} + 1 = \frac{5}{2}$

Min value $= \frac{3}{2} - 1 = \frac{1}{2}$

$-\frac{5}{2}+2 = -\frac{1}{2}$

$\boxed{-\frac{5}{2} \mid -2}\cos\left(x+\frac{\pi}{4}\right)$

$-\frac{5}{2}-2 = -\frac{9}{2}$

2

(iii) Evaluate $\lim\limits_{x\to 3}\frac{x-3}{\sqrt{x}-\sqrt{3}}$.

Example 1 (ii)

(iv) Evaluate $\lim\limits_{n\to +\infty}\left(1+\frac{4}{n}\right)^n$

12.1 Q5(v)

(iii) $\lim_{x \to 3} \frac{x-3}{\sqrt{x}-\sqrt{3}}$ $\left(\frac{0}{0}\right)$ form

$$= \lim_{x \to 3} \frac{x-3}{\sqrt{x}-\sqrt{3}} \times \frac{\sqrt{x}+\sqrt{3}}{\sqrt{x}+\sqrt{3}}$$

$$= \lim_{x \to 3} \frac{(x-3)(\sqrt{x}+\sqrt{3})}{(\sqrt{x})^2-(\sqrt{3})^2}$$

$$= \lim_{x \to 3} \frac{\cancel{(x-3)}(\sqrt{x}+\sqrt{3})}{\cancel{(x-3)}}$$

$$= \lim_{x \to 3} \sqrt{x}+\sqrt{3} = \sqrt{3}+\sqrt{3}$$

$$= 2\sqrt{3} \text{ A}$$

$$\boxed{\lim_{x \to 3} \frac{x-3}{\sqrt{x}-\sqrt{3}} = 2\sqrt{3}}$$

iv) $\lim_{n \to \infty} \left(1+\frac{4}{n}\right)^n$

$n \to \infty$

So

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = e$$

$(x^m)^p = x^{m\times p}$

$$\lim_{n\to+\infty}\left(1+\frac{4}{n}\right)^n = \lim_{n\to+\infty}\left(1+\frac{4}{n}\right)^{\frac{n}{4}\times 4}$$

$$= \left(\lim_{n\to+\infty}\left(1+\frac{4}{n}\right)^{\frac{n}{4}}\right)^4$$

$$\boxed{\lim_{n\to+\infty}\left(1+\frac{4}{n}\right)^n = e^4} \quad \text{Ans}$$

(v) Define divergent sequence.

A sequence is called divergent sequence if it does not approach to any finite real number "L" as $n \to \infty$.

i.e; $\lim_{n\to\infty} a_n \neq L$ where "L" is any finite real

$\lim_{n\to\infty}$ where "L" is any finite real no.

Examples

i) $\lim_{n\to\infty}(n^2) = \infty$

$(-1)^2 = 1$

$(-1)^3 = -1$

ii) $\lim_{n\to\infty}(-1)^n$

i) The divergent seq may increase or decrease without bound

ii) The divergent sequence may oscillate b/w different values and does not settle to any one value.

→ Example #2 → 13.1

(vi) A particle moves along a line such that its position after t hours is given by: $\boxed{s(t) = 4t^2 + 2t + 1}$ (in miles). Find the average velocity over the interval [2, 5].

(vii) Find the gradient of the curve $f(x) = 3x^2 + 2x$ at $x = 1$.

(viii) Find the derivative of $y = \frac{3}{4}x^4 + \frac{2}{3}x^3 + \frac{1}{2}x^2 + 2x + 5$ w.r.t. x.

(ix) Differentiate $\frac{2x-3}{2x+1}$ w.r.t 'x'.

Δs

0 2 5

vi) $V_{avg} = \frac{\Delta s}{\Delta t} = \frac{s(5) - s(2)}{5-2}$

$= \frac{4(25) + 10 + 1 - (16+4+1)}{3}$

$= \frac{100 + 11 - 21}{3} = \frac{90 \text{ miles}}{3 \text{ hrs}}$

111
$\underline{21}$
90

$$\boxed{\text{Average velocity} = V_{avg} = 30 \text{ miles/hour}}$$

13.1, Q8

(vii) Find the gradient of the curve $f(x) = 3x^2 + 2x$ at $x = 1$.

Let $y = f(x) = 3x^2 + 2x$

$$\frac{dy}{dx} = f'(x) = 3(2x) + 2(1)$$

$$\frac{dy}{dx} = 6x + 2$$

$$\boxed{\left.\frac{dy}{dx}\right|_{x=1} = 6(1) + 2 = 8}$$

Thus Gradient of $f(x)$ at $x = 1$ is 8

Example #7 of 13.2

(viii) Find the derivative of $y = \frac{3}{4}x^4 + \frac{2}{3}x^3 + \frac{1}{2}x^2 + 2x + 5$ w.r.t. x.

Diff w.r.t x, we get

Diff w.r.t ...

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{3}{4}x^4 + \frac{2}{3}x^3 + \frac{1}{2}x^2 + 2x + 5\right)$$

$$= \frac{3}{\cancel{4}}\cdot\left(\cancel{4}x^3\right) + \frac{2}{\cancel{3}}\cdot\left(\cancel{3}x^2\right) + \frac{1}{\cancel{2}}(\cancel{2}x) + 2$$

$$\boxed{\frac{dy}{dx} = 3x^3 + 2x^2 + x + 2} \text{ Ans}$$

13.2 Q1 (iii)

(ix) Differentiate $\dfrac{2x-3}{2x+1}$ w.r.t 'x'.

Sol: Let $y = \dfrac{2x-3}{2x+1}$

Differentiate w.r.t 'x' on both s –

by applying Quotient Theorem

$$\frac{dy}{dx} = \frac{(2x+1)\frac{d}{dx}(2x-3) - (2x-3)\frac{d}{dx}(2x+1)}{(2x+1)^2}$$

$$\frac{dy}{dx} = \frac{(2x+1)(2) - (2x-3)(2)}{(2x+1)^2}$$

$$= \frac{4x+2-4x+6}{(2x+1)^2}$$

$$\boxed{\frac{dy}{dx} = \frac{8}{(2x+1)^2}}$$ Ans

Example # 3, 14.1

(x) If $\underline{u} = 2\underline{i} + 3\underline{j} + \underline{k}$, $\underline{v} = 4\underline{i} + 6\underline{j} + 2\underline{k}$ *and* $\underline{w} = -6\underline{i} - 9\underline{j} - 3\underline{k}$, then show that $\underline{u}, \underline{v}$ *and* $\underline{w}$ are parallel to each other.

Sol:-

$$\underline{v} = 2(2\underline{i} + 3\underline{j} + \underline{k})$$

$$\boxed{\underline{v} = 2\underline{u}}$$

$\therefore$ $\underline{v}$ and $\underline{u}$ are parallel

Also,

$$\underline{w} = -3(2\underline{i} + 3\underline{j} + \underline{k})$$

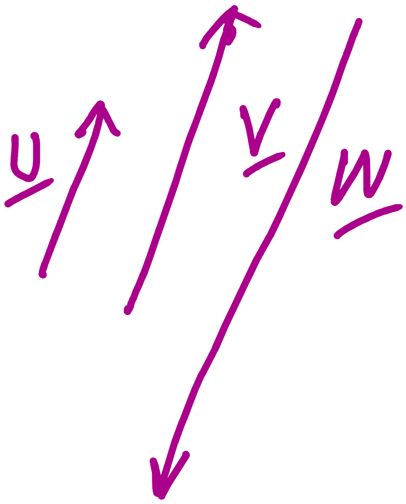
$$\boxed{\underline{w} = -3\underline{u}}$$

$\therefore$ $\underline{u}$ and $\underline{w}$ are parallel

$\therefore$ $\underline{u}$, $\underline{v}$ and $\underline{w}$ are parallel

Thus $\underline{v}$, $\underline{u}$ and $\underline{w}$ are parallel to each other.

$\underline{w} = -3\underline{u}$



14.1, Q3

(xi) Find t, so that $\left|2\underline{i}+(t-1)\underline{j}+t\underline{k}\right|=\sqrt{13}$

$$\sqrt{(2)^2+(t-1)^2+t^2}=\sqrt{13}$$

Taking square on both sides,

$$4+(t-1)^2+t^2=13$$

$$4+t^2-2t+1+t^2=13$$

$$2t^2-2t+5=13$$

$$2t^2-2t-8=0$$

$$2t^2 - 2t - 8 = 0$$

$$\boxed{t^2 - t - 4 = 0}$$

$$t = \frac{+1 \pm \sqrt{1^2 - 4(1)(-4)}}{2}$$

$$\boxed{t = \frac{+1 \pm \sqrt{17}}{2}} \text{ Ans}$$

14.2 Q13

(xii) Find the work done, if the point at which the constant force $\underline{F} = 2\underline{i} + 5\underline{j} + 3\underline{k}$ is applied to an object, moves it from $P_1(2, -3, 1)$ to $P_2(7, 5, 3)$.

Sol

Work done $= W = \underline{F} \cdot \underline{d}$ → 1

Displacement of the object $= \overrightarrow{P_1P_2} = \underline{d}$

$= (7-2)\underline{i} + (5+3)\underline{j} + (3-1)\underline{k}$

$\underline{d} = 5\underline{i} + 8\underline{j} + 2\underline{k}$

$W = (2i + 5j + 3\underline{k}) \cdot (5\underline{i} + 8\underline{j} + 2\underline{k})$

$W = (2\underline{i} + 5\underline{j} + 3\underline{k}) \cdot (5\underline{i} + 8\underline{j} + 2\underline{k})$

$W = 10 + 40 + 6$

$\boxed{\text{work done} = 56 \text{ units}}$ Ans

14.3 Q11

(xiii) Show that $|\underline{a} \times \underline{b}|^2 = |\underline{a}|^2 |\underline{b}|^2 - (\underline{a} \cdot \underline{b})^2$

$\underline{a} \times \underline{b} = |\underline{a}||\underline{b}| \sin\theta \, \underline{n} = ab\sin\theta \underline{n}$

$\underline{a} \cdot \underline{b} = |\underline{a}||\underline{b}| \cos\theta$

$LHS = |\underline{a} \times \underline{b}|^2$

$= \left| |\underline{a}| |\underline{b}| \sin\theta \, \underline{n} \right|^2$

$= |\underline{a}|^2 |\underline{b}|^2 \sin^2\theta \cdot (1)^2$

$= |\underline{a}|^2 |\underline{b}|^2 (1 - \cos^2\theta)$

$= |\underline{a}|^2 |\underline{b}|^2 - |\underline{a}|^2 |\underline{b}|^2 \cos^2\theta$

$$= |\underline{a}|^2|\underline{b}|^2 - \left(|\underline{a}||\underline{b}|\cos\alpha\right)^2$$

$$= |\underline{a}|^2|\underline{b}|^2 - (\underline{a}\cdot\underline{b})^2$$

$$= \text{RHS.}$$

Hence LHS = RHS

Thus $\boxed{|\underline{a}\times\underline{b}|^2 = |\underline{a}|^2|\underline{b}|^2 - (\underline{a}\cdot\underline{b})^2}$

1.2 Q5

Subjective (Part-II)

Note: Attempt any three questions. **$3 \times 10 = 30$**

5. (a) If $z_1 = x + yi$ and $z_2 = a + bi$, find x, y, a and b such that $z_1 + z_2 = 10 + 4i$ and $\rightarrow$ 1
$z_1 - z_2 = 6 + 2i$. $\rightarrow$ 2 **5**

(b) Solve $|x^2 - 3x + 2| > 4$ $\rightarrow$ 3.1 Q4 (vi) **5**

a) $z_1 + z_2 = 10 + 4i \rightarrow (1)$

$z_1 - z_2 = 6 + 2i \rightarrow (2)$

Adding Eqn. (1) & (2), we get

$2z_1 = 16 + 6i$

$z_1 = 8 + 3i$

$$z_1 = 8 + 3i$$

$$x + yi = 8 + 3i$$

$$\Rightarrow \boxed{x = 8}, \quad \boxed{y = 3}$$

Now subtract Eqn ② from Eqn ①, we get

$$\begin{aligned} z_1 + z_2 &= 10 + 4i \\ \underset{-}{z_1} \underset{+}{-} z_2 &= \underset{-}{6} \underset{-}{+} 2i \\ \hline 2z_2 &= 4 + 2i \end{aligned}$$

$$z_2 = 2 + i$$

$$a + bi = 2 + i$$

$$\boxed{a = 2}, \quad \boxed{b = 1}$$

Thus, $\boxed{x = 8, \ y = 3, \ a = 2, \ b = 1}$ Ans

(b) Solve $\left|x^2 - 3x + 2\right| > 4$

$$\pm\left(x^2-3x+2\right) > 4$$

$$x^2-3x+2 > 4$$

$$x^2-3x+2-4>0$$

$$\boxed{x^2-3x-2>0}$$

$$x=\frac{3\pm\sqrt{9-4(1)(-2)}}{2}$$

$$x=\frac{3\pm\sqrt{17}}{2}$$

$$x=\frac{3+\sqrt{17}}{2},\quad x=\frac{3-\sqrt{17}}{2}$$

$$x=3.56 \qquad x=-0.56$$

$$-\left(x^2-3x+2\right)>4$$

$$x^2-3x+2<-4$$

$$x^2-3x+6<0$$

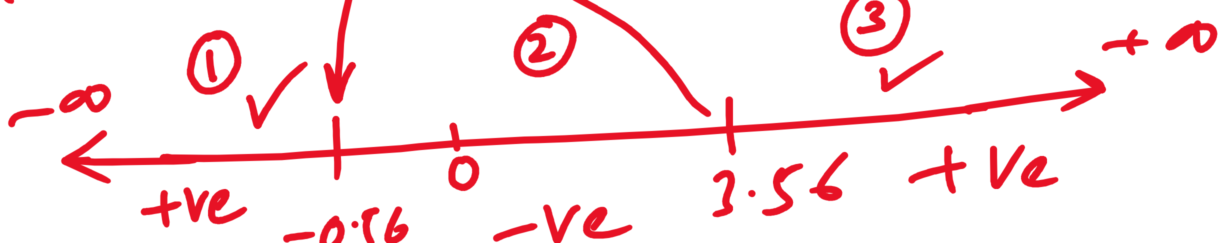
$$x=\frac{3\pm\sqrt{9-4(1)(6)}}{2}$$

$$9-24=-15<0$$

No Real Sol.

$$\text{DISC}=b^2-4ac=-15<0$$

$$\boxed{x^2-3x-2>0}$$



$$\boxed{x=\left(-\infty,\ \frac{3-\sqrt{17}}{2}\right)\cup\left(\frac{3+\sqrt{17}}{2},+\infty\right)}$$

4.2 Q3 (ix)

6. (a) Show that $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)^3$

$2c - c + a + b$

$c - 2a$

$$\begin{vmatrix} 2c & 2c & c-a-b \end{vmatrix}$$

$a-b-c-2a$ $\quad$ $2c-c+a+b$

(b) Resolve $\dfrac{x^2-10x+13}{(x-1)(x^2-5x+6)}$ into partial fractions.

→ Example # 2 5.1

a) $LHS = \begin{vmatrix} -a-b-c & 0 & 2a \\ 0 & -a-b-c & 2b \\ a+b+c & a+b+c & c-a-b \end{vmatrix}$ By $C_1 - C_3$, $C_2 - C_3$

$$= \begin{vmatrix} -(a+b+c) & 0 & 2a \\ 0 & -(a+b+c) & 2b \\ (a+b+c) & (a+b+c) & c-a-b \end{vmatrix}$$

By Taking $(a+b+c)$ as common for C_1 & C_2

$(-1)^{i+j}$

$$= (a+b+c)(a+b+c)\begin{vmatrix} -1 & 0 & 2a \\ 0 & -1 & 2b \\ 1 & 1 & c-a-b \end{vmatrix}$$

Expand by C_1

$$= (a+b+c)^2\left[-1(-c+a+b-2b)-0+1(0+2a)\right]$$

$$= (a+b+c)^2\left[c-a+b+2a\right]$$

$$= (a+b+c)^2(a+b+c)$$

$= (a+b+c)(a+b+c)$

$= (a+b+c)^3 = RHS.$

$\Rightarrow LHS = RHS$

Hence proved

$x=3 \quad x=2 \qquad x^2-5x+6 = 0$

$= x^2-2x-3x+6$

b)

(b) Resolve $\dfrac{x^2-10x+13}{(x-1)(x^2-5x+6)}$ into partial fractions.

$$\frac{x^2-10x+13}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3} \rightarrow (1)$$

$\times$ by $(x-1)(x-2)(x-3)$ on both sides of (1) we get

$$x^2-10x+13 = A(x-2)(x-3) + B(x-1)(x-3) + C(x-1)(x-2) \rightarrow (2)$$

put $\boxed{x-1=0}$ in (2) $\Rightarrow \boxed{x=1}$

$1-10+13 = A(1-2)(1-3)$

$4 = A(-1)(-2) \Rightarrow 2A = 4$

$A = \frac{4}{2} \Rightarrow \boxed{A=2}$

put $\boxed{x-2=0}$ in Eqn ② $\Rightarrow \boxed{x=2}$

$$4-20+13 = B(1)(-1)$$

$$-3 = -B \Rightarrow \boxed{B=3}$$

put $\boxed{x-3=0}$ in Eqn ②, $\Rightarrow \boxed{x=3}$

$$9-30+13 = C(2)(1)$$

$$-8 = 2C \Rightarrow \boxed{C=-4}$$

Put values of A, B & C in Eqn ①, we

$$\boxed{\frac{x^2-10x+13}{(x-1)(x-2)(x-3)} = \frac{2}{x-1} + \frac{3}{x-2} - \frac{4}{x-3}}$$

Ans

6.6 Q4 7.2 Q8

$\sqrt{ab}$

7. (a) For what value of n, $\dfrac{a^n + b^n}{a^{n-1} + b^{n-1}}$ is the positive geometric mean between a and b?

(b) ✓ How many numbers between 100 and 1000 can be formed by using digits 0, 1, 2, 3, 4, 5 without repetition? How many of them are divisible by 5?

$a, G, b \Rightarrow \boxed{G^2 = ab} \Rightarrow \boxed{G = \sqrt{ab}}$

$a, G, b \Rightarrow G^2 = ab \Rightarrow G = \sqrt{ab}$

a)

$$\frac{a^n + b^n}{a^{n-1} + b^{n-1}} = \sqrt{ab}$$

squaring both sides

$$\left(\frac{a^n + b^n}{a^{n-1} + b^{n-1}}\right)^2 = \left(\sqrt{ab}\right)^2 = ab$$

$$\frac{(a^n + b^n)^2}{(a^{n-1} + b^{n-1})^2} = ab$$

$$(a^n + b^n)^2 = ab\left(a^{n-1} + b^{n-1}\right)^2$$

$$(a^n)^2 + 2a^n b^n + (b^n)^2 = ab\left((a^{n-1})^2 + 2a^{n-1}b^{n-1} + (b^{n-1})^2\right)$$

$$a^{2n} + 2a^n b^n + b^{2n} = a^1 b^1\left(a^{2n-2} + 2a^{n-1}b^{n-1} + b^{2n-2}\right)$$

$$a^{2n} + \cancel{2a^n b^n} + b^{2n} = a^{2n-1}b + \cancel{2a^n b^n} + ab^{2n-1}$$

$$a^{2n} + b^{2n} = a^{2n-1} \cdot b + a \cdot b^{2n-1}$$

$a^{2n-1} \cdot a^{1}$

$a^{2n-1+1} - a^{2n-1} b = a b^{2n-1} - b^{2n-1+1}$

$a^{2n-1}(a - \cancel{b}) = b^{2n-1}(\cancel{a - b})$

$a^{2n-1} = b^{2n-1} \times 1$

$\Rightarrow \dfrac{a^{2n-1}}{b^{2n-1}} = 1$ $\quad \left(\dfrac{a}{b}\right)^{0} = 1$

$1^{0} = 1$, $x^{0} = 1$ $\quad \Rightarrow \left(\dfrac{a}{b}\right)^{2n-1} = 1$

$\left(\dfrac{a}{b}\right)^{2n-1} = \left(\dfrac{a}{b}\right)^{0}$

$2n - 1 = 0 \Rightarrow 2n = 1$

$\boxed{n = \frac{1}{2}}$ A

(b) How many numbers between 100 and 1000 can be formed by using digits 0, 1, 2, 3, 4, 5 without repetition? How many of them are divisible by 5?

501 $\quad$ H T U $\quad$ 2 3 $\quad$ 023

101 $\quad$ Total Numbers b/w 100 & 1000

111

111 Numbers b/w

$= \underset{H}{5} \times \underset{T}{5} \times \underset{U}{4} \rightarrow$ Counting Principle

101
100 → 1000

102
103
104
105
110
120

$= \boxed{100 \text{ numbers}}$

5
10
15
20

0, ~~1~~, 2, 3, 4, 5

When

ii) Unit's place = 0

$\underset{H}{5} \times \underset{T}{4} \; \boxed{\underset{U}{0}} \rightarrow$ Fixed

$= 20$ numbers

If Unit's place $= 5$

~~0~~, 1, 2, 3, ~~4~~

$\underset{H}{4} \times \underset{T}{4} \; \boxed{5}$

$= 16$ numbers

Total 3 digits numbers divisible by 5 are $20 + 16 = \boxed{36}$

10.4 Q5(ii)

8. (a) Prove that $\sin\frac{\pi}{9}\sin\frac{2\pi}{9}\sin\frac{\pi}{3}\sin\frac{4\pi}{9} = \frac{3}{16}$

8. (a) Prove that $\sin\frac{\pi}{9}\sin\frac{2\pi}{9}\sin\frac{\pi}{3}\sin\frac{4\pi}{9} = \frac{3}{16}$

(b) Given the function $f(x)=\begin{cases}2x+3, & x\le 1\\ -x+4, & x>1\end{cases}$ ↗ 12.2 Q7

$\frac{180}{9}=20^\circ$

Discuss the limit and continuity at $x = 1$.

a) $L.H.S = \sin\frac{\pi}{9}\sin\frac{2\pi}{9}\sin\frac{\pi}{3}\sin\frac{4\pi}{9}$

$= \sin 20^\circ \sin 40^\circ \sin 60^\circ \sin 80^\circ$

$= \frac{\sqrt{3}}{2}\sin 20^\circ \sin 40^\circ \sin 80^\circ$

$= -\frac{1}{2}\times\frac{\sqrt{3}}{2}\sin 40^\circ\left(-2\sin 80^\circ \sin 20^\circ\right)$

As we know that $-2\sin\alpha\sin\beta = \cos(\alpha+\beta)-\cos(\alpha-\beta)$

$= -\frac{\sqrt{3}}{4}\sin 40^\circ\left(\cos 100^\circ - \cos 60^\circ\right)$

$= -\frac{\sqrt{3}}{4}\sin 40^\circ\left(\cos 100^\circ - \frac{1}{2}\right)$

$= -\frac{\sqrt{3}}{4}\left[\cos 100^\circ \sin 40^\circ - \frac{1}{2}\sin 40^\circ\right]$

$\ldots(\alpha+\beta) - \sin(\alpha-\beta)$

7 (

As we know that $2\cos\alpha\sin\beta = \sin(\alpha+\beta) - \sin(\alpha-\beta)$

$$= -\frac{\sqrt{3}}{4}\left[\frac{1}{2}\left(\underline{\underline{2\cos 100^\circ \sin 40^\circ}}\right) - \frac{1}{2}\sin 40^\circ\right]$$

$$= -\frac{\sqrt{3}}{4}\times\frac{1}{2}\left[2\cos 100^\circ \sin 40^\circ - \sin 40^\circ\right]$$

$$= -\frac{\sqrt{3}}{8}\left[\sin 140^\circ - \sin 60^\circ - \sin 40^\circ\right]$$

$$= -\frac{\sqrt{3}}{8}\left[\sin 140^\circ - \sin 40^\circ - \frac{\sqrt{3}}{2}\right]$$

S A T C

$$= -\frac{\sqrt{3}}{8}\left[\sin(\underbrace{180^\circ - 40^\circ}_{II}) - \sin 40^\circ - \frac{\sqrt{3}}{2}\right]$$

$$= -\frac{\sqrt{3}}{8}\left[\cancel{\sin 40^\circ} - \cancel{\sin 40^\circ} - \frac{\sqrt{3}}{2}\right]$$

$$= -\frac{\sqrt{3}}{8}\times\left(-\frac{\sqrt{3}}{2}\right)$$

$$= \frac{(\sqrt{3})^2}{16} = \frac{3}{16} = R.H.S$$

Hence

$= 16$

LHS = RHS. Hence proved

(b) Given the function $f(x)=\begin{cases} 2x+3, & x\le 1 \\ -x+4, & x>1 \end{cases}$

Discuss the limit and continuity at $x = 1$.

Sol: To check whether $\lim_{x\to 1}$ exists or not

$$\lim_{x\to 1} f(x) = \lim_{x\to 1^-} f(x) = \lim_{x\to 1^+} f(x)$$

$$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} (2x+3) = 5$$

$$\lim_{x\to 1^-} f(x) = 5 \longrightarrow (1)$$

$$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} (-x+4) = -1+4 = 3$$

$$\lim_{x\to 1^+} f(x) = 3 \longrightarrow (2)$$

As $\lim_{x\to 1^-} f(x) \neq \lim_{x\to 1^+} f(x)$

$\therefore \lim f(x)$ does not exist,

So, $\lim_{x\to 1} f(x)$ does not exist,

Continuity $f(1) = \lim_{x\to 1} f(x)$

The $f(x)$ is discontinous at $x=1$
as $\lim_{x\to 1} f(x)$ does not exist.

13.2 Q3

9. (a) Differentiate $\dfrac{(\sqrt{x}+1)(x^{\frac{3}{2}}-1)}{x^{\frac{3}{2}}-x^{\frac{1}{2}}}$ with respect to x.

14.2 Q3

(b) If $|\underline{a}|=3$, $|\underline{b}|=4$ and $|\underline{a}+\underline{b}|=5$. Find the angle between $\underline{a}$ and $\underline{b}$.

a) Let $y = \dfrac{(\sqrt{x}+1)(x^{3/2}-1)}{x^{3/2}-x^{1/2}}$

$x^{\frac{3}{2}} = x^1 \cdot x^{\frac{1}{2}}$

$$y = \frac{(\sqrt{x}+1)\left((x^{1/2})^3-(1)^3\right)}{x\cdot x^{\frac{1}{2}} - x^{\frac{1}{2}}}$$

$$y = \frac{(\sqrt{x}+1)(\sqrt{x}-1)(x+\sqrt{x}+1)}{x^{\frac{1}{2}}(x-1)}$$

$$x^{\frac{1}{2}}(x-1)$$

$$y = \frac{\cancel{(x-1)}(x+\sqrt{x}+1)}{\sqrt{x}\,\cancel{(x-1)}}$$

$$y = \frac{x+\sqrt{x}+1}{\sqrt{x}}$$

$x^{1-\frac{1}{2}}$

$$y = \frac{x^1}{\sqrt{x}} + 1 + \frac{1}{\sqrt{x}}$$

$$y = x^{\frac{1}{2}} + 1 + x^{-\frac{1}{2}}$$

Diff. w.r.t x on both sides

$$\frac{dy}{dx} = \frac{1}{2}x^{\frac{1}{2}-1} + 0 - \frac{1}{2}x^{-\frac{1}{2}-1}$$

$$= \frac{1}{2}x^{-\frac{1}{2}} - \frac{1}{2}x^{-\frac{3}{2}}$$

$$= \frac{1}{2x^{1/2}} - \frac{1}{2x^{3/2}}$$

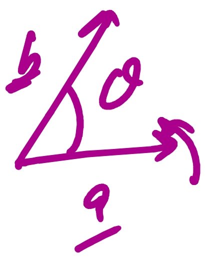
$$= \frac{1}{2x^{1/2}} - \frac{1}{2x \cdot x^{1/2}}$$

$$= \frac{x - 1}{2x \cdot x^{1/2}}$$

$$\boxed{\frac{dy}{dx} = \frac{x-1}{2x\sqrt{x}} = \frac{x-1}{2x^{3/2}}}$$

(b) If $|\underline{a}| = 3$, $|\underline{b}| = 4$ and $|\underline{a} + \underline{b}| = 5$. Find the angle between $\underline{a}$ and $\underline{b}$.

Sol Given $|\underline{a} + \underline{b}| = 5$



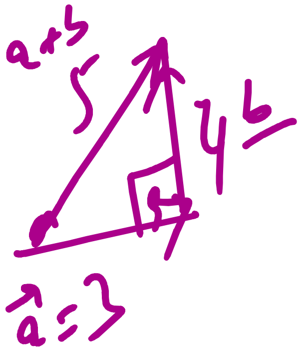
Squaring both side

$$|\underline{a} + \underline{b}|^2 = 5^2$$

$$|\underline{a}|^2 + |\underline{b}|^2 + 2|\underline{a}||\underline{b}|\cos\theta = 25$$

$$9 + 16 + 2(3)(4)\cos\theta = 25$$

$$24\cos\theta = 0$$



$$24 \cos\theta = 0$$

$$\cos\theta = 0$$

$$\theta = \cos^{-1}(0)$$

$$\boxed{\theta = 90^\circ}$$